Syllabus
BSc Faculty of Computer Science and Engineering

Subject	**Web Design and Development**			
	Type	**Semester**	**ECTS**	**Code**
	OBLIGATIVE (O)	3	5	40WEN255
Course Lecturer **Course Assistant** **Course Tutor**	Greta Ahma, can.PhD.			
Aims and Objectives	The aim of this course is to introduce the methods and techniques used in Web Design and Development. This course draws upon previous programming experience to develop practical web development and maintenance skills. The students also will get knowledge of some well-known web technologies and tools for developing web applications: mark-up languages, styling, client and server side programming.			
Prerequisite(s)	This is a follow up course that requires an introduction to programming language.			
Learning Outcomes	By the end of this course students should be able to: • Identify web technologies and techniques for developing web pages & simple web application. • Develop user-interfaces for web applications. • Implement front-end technologies for UI design and implementation. • Implement responsive layout of web page • Manage data and user type controls and validation • Implement OOP server-side technology for basic CRUD operations • Develop a simple full-stack webpage with high-performance, secure and accessible website, upon a client's request.			
15 Week Course Content	**Course Plan** **Unit I**: *introduction to web technologies and it's basic architecture* Overview of Course, web technologies and web architecture **Unit II:** *Webpage design and responsive layout* HTML5 and CSS3, CSS Selectors, Pseudo-classes, Responsive Web Design & Layouts, Flexbox, User Experience and Design **Unit III:** *Implement Dynamic Behaver and interaction with user's perspectives* Introduction to Client-Side Scripting (JavaScript), JS Function, Objects, Understanding and Modifying the DOM with JS, JS Event Handlers, JS Form Validation **Unit IV**: *basic concepts of Back-end Development* Server-Side programming: Basic concepts PHP (Setup, syntax, variable types and structure controls), PHP Superglobals, Sessions and Cookies, Form Handling. **Unit V:** *Server-side technologies & data manipulation (basic CRUD Operation)*			

	Introduction to OOP with PHP including (Visibility, Inheritance, Interface, etc) or other server-side web technologies, Database connection & interaction (MySQL/Relational databases) including implementation of CRUD operations

Teaching/Learning Methods

Teaching/Learning Activity	Weight (%)
Lectures	30%
Seminars	0%
Group project	20%
Laboratory	30%
Self-study	20%

Assessment Methods

Assessment Activity	Number	Week	Weight (%)
1. Group project (max 2 students)	2	7,15	40%
2. Mid-term Test	0		0%
3. Finale – Test	1	15	60%
4. Attendance (student are required to attend min 75% lecture classes and laboratory exercises to be qualified to take exams)			0%
5. Students from the oldest generation are allowed to take the exam with full weight (100%).			
6. Course activities such as project, assignments or homework's are obligator			

Course resources

Resources	Number
1. Lectures hall (e.g)	1
2. Laboratory exercises (e.g)	1
3. Moodle platform	
4. Software: Git, Text Editors (notpad++, Visual Studio Code), Photoshop, WAMP Server, XAMPP	
5. Projector	

ECTS Workload

Activity	Weekly hrs	Total workload
1. Lectures	2	30
2. Seminars/Projects		24
3. Laboratory	2	30
4. Self-study		78
5. Finale Test		

Literature/References

1. HTML and CSS: Design and Build Websites - Jon Duckett
2. JavaScript and JQuery: Interactive Front-End Web Development – Jon Duckett
3. Learning PHP, MySQL & JavaScript: With jQuery, CSS & HTML5 (4th Edition) - Robin Nixon
4. Modern PHP, New Features and Good Practices - Josh Lockhart

	5. Web Engineering: A Practioner's Approach - Roger S. Pressman
Contact	Email: greta.ahma@ubt-uni.net. Consultation: Monday 15:30-17:00